PAVEL CHERKASHIN

Software Engineer • Agent Harness & Distributed Systems • Managed Agents Platform

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Operating Focus: *"Architecting the runtime platform behind real-time multimodal voice agents. Bridging low-latency transport engineering, autonomous agent harnesses (fleetd daemon, slot concurrency, zero-trust git handoffs), and automated eval suites (LLM-as-judge scoring, pipeline tracing, latency regression catching) to deliver sub-second, resilient voice agent deployments at production scale."*

CORE TECHNICAL CAPABILITIES & AGENT HARNESS SUBSTRATE

- **Agent Harnesses & Daemon Runtimes:** Custom multi-agent orchestrator architecture (fleetd Rust daemon), deterministic slot scheduling (WIP=1), proactive backlog vacuuming, zero-trust git handoffs, and process isolation.
  - **Conversational Latency Optimization:** End-to-end tracing across STT → LLM → TTS pipelines; profiling Time-To-First-Token (TTFT), streaming chunk delivery, socket buffers, and turn-taking latency.
  - **Eval Suites & LLM-as-Judge Scoring:** Automated synthetic evaluation harnesses, semantic regression tracking, LLM-as-judge scoring pipelines, and formal verification substrates (Lean 4/Mathlib) for behavioral safety.
  - **Deployment Resilience & Systems Scale:** High-throughput streaming backends, zero-downtime hot reloading, cold-start reduction, circuit breakers, and fault-tolerant distributed queuing architectures.
  - **Frontier Coding-Agent Velocity:** Fluent daily operation across modern agentic loops (Claude Code, Gemini CLI, OpenAI tools); disciplined spec-driven development, sharp PR reviews, and automated CI/CD gates.

SYSTEMS ENGINEERING & AGENT HARNESS EXPERIENCE

Mindrock Autonomous Agent Mesh — *Principal Systems Architect & Harness Builder* | 2024 – Present

- **fleetd Daemon & Slot Concurrency:** Engineered the monolithic fleet daemon in Rust, coordinating distributed agent slots across macOS nodes (wolfram, darwin) with autonomous task-vacuum loops, rate-limiting, and strict WIP caps.
  - **Protocol Phi & Zero-Trust Coordination:** Created sealed JSON envelope transport (proto: phi-0.1) across Git repositories, enforcing cryptographically verifiable handoffs, isolated workspace lifecycles, and tamper-proof SHA-256 receipts.
  - **Continuous Eval & QA Gate Architecture:** Built automated multi-tier QA test suites (cross-agent verification between Gemini, Codex, and Claude) catching layout, functional, and semantic regressions prior to human gate approval.
  - **Real-Time Event Telemetry:** Deployed structured telemetry pipes capturing turn latency, agent error recovery, and tool-call execution metrics, identifying runtime contention and latency regressions across distributed nodes.

Vestor.in & High-Throughput Distributed Platforms — *Founding Architect* | 2018 – 2024

- **Event-Driven Distributed Engines:** Architected fault-tolerant streaming backends and transaction processing engines operating with low-latency streaming bounds under burst traffic loads.
  - **Resilient Transport & Gateway Layer:** Implemented distributed webhook and websocket streaming dispatchers with automatic retry exponential backoff, jittered health checks, and zero-downtime rolling deploys.
  - **Telemetry & Regression Profiling:** Built distributed tracing pipelines identifying database locking and network queue contention, significantly reducing tail (p99) latency and operational jitter.

Enterprise Systems & Technology Leadership — *Siebel Systems, Sputnik Labs, Microsoft* | Prior

- Country Manager at Siebel Systems; Founder of Sputnik Labs (scaled to 500+ engineers executing mission-critical distributed systems and CRM integrations for telecom and banking leaders).
  - Regional Director at Microsoft; authored authoritative university textbook on enterprise software and automated customer architectures; mentored hundreds of engineers on reliable system design.

ROLE ALIGNMENT, TRANSPARENT GAPS & OPERATING POSTURE

- **Intentional IC Backend Focus:** Experienced founder/operator deliberately pursuing an individual contributor backend systems role to build foundational, high-leverage agent runtime infrastructure for Cartesia's Managed Agents.
  - **First-Principles Ramp on Voice Protocols:** Direct mastery of agent harnesses, distributed queuing, and streaming sockets; transparently ramping on low-level WebRTC/SIP/RTP telephony protocols and voice frameworks (Pipecat/LiveKit).
  - **Engineering Discipline & Collaborative Presence:** Deeply committed to in-person San Francisco collaboration, sharp technical specifications, rigorous peer design reviews, and high-velocity shipping culture.

EDUCATION & SYSTEMS FOUNDATION

- **B.S. / M.S. Applied Mathematics & Computer Science:** Russian State University of Oil & Gas (Quantitative algorithms, discrete mathematics, network theory, and systems programming).